A bisector on a semicircle

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Let Γ be a semicircle on a line ℓ, and C and D points of Γ. Let A and B be the intersections of the tangents at C and D with ℓ, and assume that A and B lie on either side of Γ. Let E be the intersection of (AC) and (BD), and F the projection of E on ℓ. Show that the line (FE) bisects $\widehat{CFD}$.